

Intermediate Hands-On Tasks — Linux Group Management

Task 1: Multi-Team Project Access Control

Scenario

You are managing a server with three teams:

- Backend team
- Frontend team
- QA team

Requirements

1. Create groups:
 - `backend`
 - `frontend`
 - `qa`
 2. Create users:
 - `dev1`, `dev2` → backend
 - `ui1`, `ui2` → frontend
 - `qa1` → qa
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Task 2: Change Group Name Without Breaking Production

Scenario

Company renames:

`backend` → `api-team`

Requirements:

1. Rename group
 2. Ensure:
 - Users remain in group
 - GID remains unchanged
 - Files still show correct group name
 3. Verify no file ownership breaks
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Task 3: Change GID in Production (Advanced Risk Handling)

Scenario

Security team requires GID change:

`api-team` GID must change from 1050 → 3000

Tasks:

1. Change GID
2. Identify files owned by old GID
3. Fix ownership properly
4. Validate everything

⚠ This is a real DevOps-level task.

Task 4: Implement Group Admin Model (Legacy Scenario)

Scenario

You want `dev1` to manage backend team members without `sudo`.

Tasks:

1. Create group `release`
 2. Set `dev1` as group admin
 3. Switch to `dev1`
 4. Add `dev2` to `release`
 5. Remove `dev2`
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Real-World Scenario: Project-Based Access Control in a Growing Company

Background

Your company is starting a new internal project called:

Phoenix

The CTO wants:

- Only project members should collaborate.

- Developers may move in and out of the project.
- A team lead should manage members without full root access.
- No one outside the project should be accidentally added.
- The project may later be renamed.

You are responsible for setting up and managing this using **Linux groups only**.

Your Tasks

Step 1: Create the Project Group

Create a group:

phoenix

Verify it exists using:

getent group phoenix

Step 2: Add Initial Project Members

Users in system:

- dev1
- dev2
- qa1
- manager1

Add:

- dev1
- dev2
- qa1

as **secondary** group members.

Verify membership using:

```
id dev1
groups dev2
```

✓ Step 3: Make Project Lead a Group Administrator

The CTO wants **manager1** to manage project members **without sudo**.

Set **manager1** as group administrator.

Then:

1. Switch to **manager1**
2. Add a new user **dev3** to phoenix group
3. Remove **qa1** from phoenix group

Verify everything works without root.

✓ Step 4: Simulate Employee Exit

Now assume:

- **dev2** leaves the project.

Tasks:

1. Remove `dev2` from the phoenix group.
 2. Confirm they are no longer a member.
 3. Ensure their primary group remains unchanged.
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✅ Step 5: Project Renaming (Business Change)

Management decides to rename the project:

`phoenix` → `phoenix-app`

Tasks:

1. Rename the group.
2. Ensure:
 - GID remains same.
 - All members remain.
3. Verify new group name using:

```
getent group phoenix-app
```

✅ Step 6: Audit Task (Very Important in Real World)

Now generate answers for:

1. List all members of phoenix-app.
2. Check which users belong to more than 2 groups.
3. Identify groups that currently have no members.
4. Confirm who is group administrator.

You must do this using:

- `/etc/group`
 - `/etc/gshadow`
 - `getent`
 - `id`
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